

An HCN(1-0) and 3 mm continuum study of the circumnuclear disk of NGC 4429: dense-gas mass limits and an unresolved verdict on the central molecular hole

ALMA THESIS PLANNER (AUTOMATED PIPELINE)¹

¹*Automated analysis pipeline output – not peer reviewed*

ABSTRACT

The lenticular galaxy NGC 4429 hosts a well-studied circumnuclear molecular disk with a reported central hole in CO emission around its supermassive black hole, raising the question of whether the nucleus is genuinely gas-poor or merely faint in the tracers observed so far. We used ALMA Band 3 observations — a 126-channel HCN(1-0) cube ($1.88'' \times 1.81''$ beam, corresponding to ~ 150 pc at our adopted distance of 16.5 Mpc; 49.5 km/s channels) and a wideband 3 mm continuum image ($1.63'' \times 1.55''$ beam) — to characterize the dense gas and nuclear continuum in the inner few hundred parsecs. We measured a per-channel RMS of 0.848 mJy/beam in the cube and searched ± 300 km/s around the systemic velocity of 1104 km/s in nested apertures of one, two, and three beam radii centered on the nucleus. HCN(1-0) was not detected in any aperture: the central 147 pc aperture yielded a 3σ integrated-flux upper limit of 0.765 Jy km/s, with limits of 1.515 and 2.261 Jy km/s at 295 and 442 pc, corresponding to dense-gas mass upper limits of 8.7×10^6 , 1.7×10^7 , and 2.6×10^7 solar masses for a conversion factor of 10 Msun $(\text{K km/s pc}^2)^{-1}$. Because both the central beam and the surrounding one-to-three-beam annulus (3σ limit 2.128 Jy km/s) are non-detections, the data cannot distinguish a genuine central cavity from a nucleus that is uniformly faint in HCN(1-0), and we report the cavity question as unconstrained at this sensitivity and resolution. In contrast, the 3 mm continuum was clearly detected at the nuclear position, with 0.927 ± 0.084 mJy ($S/N = 11$) in the central beam and consistent fluxes in larger apertures, indicating a compact source. These measurements place the first HCN(1-0)-based ceilings on the dense-gas reservoir within ~ 150 –440 pc of the black hole in NGC 4429 and establish the sensitivity that deeper or higher-resolution observations must exceed to settle whether its molecular hole is truly devoid of dense gas.

1. INTRODUCTION

The centers of galaxies are where the interplay between supermassive black holes and their host interstellar medium is most direct, and yet they are among the hardest places to characterize observationally. Circumnuclear molecular disks — rotating structures of cold gas on scales of tens to hundreds of parsecs — serve both as potential fuel reservoirs for nuclear activity and as dynamical tracers of the gravitational potential close to the black hole. Whether such a disk extends all the way to the nucleus, or instead terminates in a central cavity, bears on how gas is delivered to (or expelled from) the immediate vicinity of the black hole, and on the physical mechanisms — tidal disruption of clouds, feedback from a radio nucleus, resonances in the stellar potential, or simple consumption — that shape the innermost gas distribution.

The lenticular galaxy NGC 4429, a member of the Virgo cluster, is a particularly clean laboratory for this question. It hosts a well-studied, regularly rotating

circumnuclear molecular disk, and observations of CO emission have revealed an apparent hole in the molecular gas distribution at its center, coincident with the supermassive black hole. The straightforward interpretation is that the nucleus is genuinely evacuated of molecular gas. But a single-tracer deficit admits a second reading: CO traces the bulk of the molecular medium at moderate densities, and its emission can be suppressed near a galactic nucleus by excitation effects, changes in optical depth, or chemistry, even where substantial gas remains. In the dense, hot, and possibly irradiated environment within a hundred parsecs of a black hole, the gas that survives may be preferentially dense — precisely the phase to which CO at low rotational transitions is least specifically sensitive. Distinguishing a true cavity from a tracer artifact therefore requires an independent probe of the gas, and lines of high-dipole-moment molecules such as HCN, whose low-lying transitions require densities orders of magnitude higher than CO to excite, are the natural choice.

The difficulty is that this test is demanding in exactly the ways that make it easy to over-interpret. Dense-gas lines are intrinsically faint, typically an order of magnitude or more below CO, so sensitivity limits the depth of any statement. The reported central hole in NGC 4429 spans tens of parsecs, while achievable beam sizes at millimeter wavelengths for a Virgo-distance target are often larger; when the cavity is smaller than the beam, the central measurement mixes flux from the cavity with flux from the surrounding disk, and beam convolution can wash out even a total central deficit. Compounding this, an absence of detected line emission is inherently ambiguous: it is consistent both with a hole in an otherwise bright disk and with a disk that is uniformly faint in the chosen tracer. Any honest analysis must therefore state, in advance, what configurations of gas the data can and cannot distinguish — and must be prepared to report “unconstrained” as a legitimate answer.

This paper describes a planned study designed with those limitations built in from the start. The inputs are two science-ready ALMA Band 3 products for NGC 4429: a 126-channel cube covering the HCN(1-0) line at approximately 1.9 arcsecond resolution — about 150 parsecs at the adopted Virgo distance — with roughly 50 km/s channels, and a wideband 3 mm continuum image at slightly finer resolution. From these two files alone, the study will characterize the noise properties of the data, extract spectra in nested apertures anchored to the nucleus over a generous velocity window around the systemic velocity, and apply a pre-registered detection criterion that selects exactly one analysis branch: spatial and spectral characterization of the dense gas if the line is detected, or a rigorous program of aperture-integrated flux upper limits and spectral stacking if it is not. In either branch, the measured fluxes or limits will be converted to dense-gas masses using standard conversion factors, with the factor-of-several systematic uncertainty in that conversion propagated explicitly so that every mass is reported as a bracketed range rather than a single number. In parallel, the continuum image will be searched for a compact nuclear 3 mm source, whose presence, flux, and extent provide independent context on nuclear activity.

A deliberate feature of the design is that the central-cavity question is treated as a bounded constraint problem rather than a measurement. Because the known hole is smaller than the beam, no attempt is made to image it; instead, simple parametric disk models — filled and holed, over a bracketed family of surface-brightness profiles — will be convolved with the actual synthesized beam and compared against the observed central-beam flux and radial profile. A cavity configuration will be

declared excluded only if it fails this comparison under every profile assumption in the bracket; anything excluded under some assumptions but not others will be labeled unconstrained. The intended deliverable is a quantitative statement of the form “cavities below a given radius or shallower than a given depth are unconstrained at this resolution,” with the boundary values determined by the data rather than by expectation.

Success for this study is defined by the reliability of its constraints, not by any particular astrophysical outcome. Verification is built into the plan at several levels: the detection decision is made by a pre-registered numerical criterion recorded before branch-specific analysis begins; the convolution machinery used for the cavity constraints is validated by flux-conservation checks against known inputs; independent flux measurements (integrated spectra versus moment maps, aperture photometry versus model fits for the continuum) are cross-compared and required to agree within their uncertainties; and the entire pipeline is re-run end-to-end from the two input files to confirm that every reported number regenerates. Whatever the data yield — a detection of dense gas, upper limits on its mass, or a verdict of “unconstrained” on the cavity — the result will be the first HCN(1-0)-based characterization of the dense-gas reservoir in the inner few hundred parsecs of NGC 4429, and it will establish concretely the sensitivity and resolution that future observations must exceed to settle whether the molecular hole around this black hole is truly empty.

2. METHODS

2.0.1. Data

All measurements in this work were made on two ALMA Band 3 imaging products for NGC 4429: a primary-beam-corrected wideband continuum image and a primary-beam-corrected 126-channel spectral cube covering the HCN(1-0) transition (rest frequency 88.6316 GHz). Both files were downloaded as FITS products and every analysis step described below was executed directly against them; no quantity in this paper derives from any other input data. The continuum image has a synthesized beam of $1.63'' \times 1.55''$ at position angle 80.4° , and the cube a beam of $1.88'' \times 1.81''$ at position angle 79.8° , with a channel width of 14.65 MHz (49.5 km/s in the radio convention at the line frequency). We adopted a distance of 16.5 Mpc (Virgo Cluster mean, NGVS SBF catalog) and a heliocentric systemic velocity of 1104 km/s; the same distance was used consistently in every physical conversion.

2.0.2. Software and reproducibility

The analysis was carried out in a single version-controlled Python environment using *astropy* (FITS

I/O, world coordinates, units, and constants), spectral-cube and `radio_beam` (cube handling and beam objects), `numpy` and `scipy` (numerics, convolution, and statistics), `photutils` (aperture photometry), and `matplotlib` (figures). Every number reported in this paper was produced by a script in the project repository: cube utilities, photometry, and the model-convolution machinery were implemented as separate modules, each writing its figures and a machine-readable table of measured quantities to a results directory. The full pipeline was re-run end-to-end from the two FITS files with a single driver script at the assembly stage, and all reported values were confirmed to regenerate.

2.0.3. Data verification and noise characterization

Before any science measurement, both FITS headers were audited and the synthesized beam parameters, pixel scale, spectral-axis convention, brightness unit, and reference coordinates were recorded. The spectral axis of the cube was converted to velocity relative to the HCN(1-0) rest frequency using the radio convention, and we verified that the redshifted line at the adopted systemic velocity falls well inside the 126-channel window, recording the corresponding channel range. The line search was conducted over ± 300 km/s around the systemic velocity, corresponding to 13 channels. The field center was checked against the ICRS nuclear position RA = 12:27:26.503, Dec = +11:06:27.58, and the pixel coordinates of that position were located in both images; all apertures were anchored to this position.

Because both products are primary-beam corrected, the noise rises toward the field edge, so the RMS was measured in the inner $\sim 30''$ only, where the correction is mild. For the cube, the per-channel RMS was computed across all 126 channels from emission-free channels using robust statistics (median absolute deviation scaled to an equivalent Gaussian sigma), and the median per-channel RMS was adopted as the sensitivity reference for all subsequent calculations. The continuum RMS was measured the same way from emission-free sky regions. Spatially integrated spectra were extracted in circular apertures centered on the nucleus, converting Jy/beam to Jy by dividing by the number of pixels per beam, with per-channel uncertainties propagated from the measured noise and scaled by the number of independent beams in each aperture.

2.0.4. Beam-convolution sanity checks

To establish the detectability floor at this resolution, we built a toy-model convolution framework before making the branch decision. Simple model disks were generated on the cube's pixel grid — a filled exponential

disk and the same disk with a central cavity of specified radius and fractional depth — and each was convolved with the actual synthesized beam, constructed as a Gaussian kernel from the header beam parameters via `radio_beam` and convolved with `astropy/scipy` convolution routines. Comparing the central-beam flux of holed versus filled models quantified how large and how deep a cavity must be to produce a measurable central deficit at $\sim 1.9''$ resolution given the measured RMS. The same code was verified to conserve total model flux under convolution to better than 1%.

2.0.5. Detection criterion and branch decision

A pre-registered detection criterion was applied by script: the analysis would follow the detection branch if either (a) the integrated line flux over the expected velocity window reached at least 5σ , or (b) any spatial position showed per-beam peak signal-to-noise of at least 3 in at least 3 contiguous channels. Both tests were computed and the numbers recorded in a dated gate memo. The criterion was not met, so the non-detection branch — upper limits and aperture-integrated constraints — was the branch executed, and all subsequent line tooling was built for that branch alone.

2.0.6. Line photometry and upper limits

Integrated HCN(1-0) fluxes were measured by beam-scaled aperture photometry: circular apertures of radius 1, 2, and 3 beam widths ($1.84''$, $3.68''$, and $5.52''$, corresponding to 147, 295, and 442 pc at the adopted distance), plus a 1–3 beam annulus, all centered on the nuclear position. In each aperture the flux was integrated over the 13-channel search window, with the uncertainty computed from the per-channel RMS as $\sigma_{\text{chan}} \times \Delta v \times \sqrt{N_{\text{chan}}}$, scaled by the square root of the number of independent beams in the aperture. Measurements below the detection threshold were reported as 3σ upper limits on the integrated line flux. Per-beam limits were also tabulated for assumed line widths of 50, 100, and 200 km/s (1, 2, and 4 channels), and a map of the per-pixel 3σ limit across the inner disk was produced to document the spatial variation imposed by the primary-beam correction. Aperture spectra were stacked at zero velocity shift with inverse-variance weighting, and the RMS of each stacked spectrum was measured directly from its line-free channels rather than taken from the propagated value; at 49.5 km/s channels, rotation moves emission across only ~ 1 –3 channels over most of the disk, so this plain stack captures nearly all of the available sensitivity gain, which we quantified with a comparison of the channel width against the plausible projected rotation spread. The velocity-coherent shift-and-stack

stretch goal was not pursued, and the reported limits are the plain-stack limits — the pre-committed fallback.

2.0.7. *Kinematic check*

As a consistency check on any low-level emission, all spatial pixels exceeding a 3σ per-channel threshold were identified and an intensity-weighted (moment-1) velocity was computed over those pixels, to test whether their velocity distribution brackets the systemic velocity as expected for a rotating disk. A position-velocity cut was extracted along the apparent elongation of the emission for the same single purpose — verifying a rotation-like velocity gradient — with the beam and channel width drawn on the panel. No kinematic fitting, rotation-curve extraction, or black-hole mass modeling was attempted: the resolution and the coarse channelization do not support it. For the same reason no moment-2 map was computed, as at this channel width it would be dominated by channelization.

2.0.8. *Continuum measurement*

On the wideband Band 3 continuum image, the RMS was measured as above and the nuclear position was tested for a source. Aperture photometry was performed in the same beam-scaled apertures used for the line (radii of 1, 2, and 3 beams), with uncertainties scaled by the number of independent beams per aperture, and each measurement was classified as a detection or a 3σ limit against the local noise. Where the nuclear source exceeded the 5σ threshold, its photometry was compared against the ICRS nuclear coordinates, and the same photometry over the larger circumnuclear-disk apertures constrained any extended 3 mm emission. The nuclear continuum flux density was converted to a monochromatic luminosity at the adopted 16.5 Mpc distance.

2.0.9. *Central-depression diagnostic*

To test for a central dense-gas deficit, we compared the central 1-beam aperture against the 1–3 beam annulus using a single pre-defined metric: the base-10 logarithm of the ratio of the central-aperture integrated flux (or limit) to the annulus flux (or limit). The toy-model convolution machinery described above provided the interpretation framework: model disk surface-brightness profiles of two functional forms (exponential and Gaussian) over a bracketed range of scale lengths, each with a central cavity of radius on a grid from ~ 10 pc to ~ 1.5 beam FWHM and fractional depth from 0 to 100%, were normalized to the measured aperture limits, convolved with the true beam, and compared to the observed central-beam constraint. A cavity model was considered excluded only if it violated the measurement by more than

3σ under every disk profile in the bracket; cells excluded under one profile assumption but not another were labeled unconstrained. Because both the central aperture and the annulus proved to be non-detections at the achieved sensitivity, the diagnostic was reported with the explicit caveat that the data cannot distinguish a central cavity from an overall HCN-faint nucleus.

2.0.10. *Line luminosity and dense-gas mass*

Each integrated HCN(1-0) flux or limit was converted to a line luminosity L'_{HCN} in K km/s pc² using the standard relation between velocity-integrated flux, observed line frequency, and luminosity distance, at the adopted 16.5 Mpc. Dense-gas masses were obtained from L'_{HCN} using $\alpha_{\text{HCN}} = 10.0 \text{ M}_{\text{sun}}/(\text{K km/s pc}^2)$ (Gao & Solomon 2004), with the factor-of-several literature range in α_{HCN} treated as a bracketed systematic quoted separately from the statistical error. Masses (or limits) were computed for each beam-scaled aperture. A published CO flux could not be defensibly matched to the HCN apertures in resolution and extent within the allotted effort, so, following the pre-committed fallback, the HCN dense-gas constraints are reported alone rather than as an HCN/CO dense-gas fraction.

2.0.11. *Internal cross-checks*

At assembly, the following cross-checks were performed and recorded: agreement between aperture-integrated fluxes and the integrated-spectrum measurements; agreement between aperture photometry and fitted fluxes for the continuum; flux conservation of the toy-model convolution to better than 1%; and consistency between the gate-memo numbers and the final measured values. All checks passed under the single-driver re-run of the pipeline from the two input FITS files.

3. RESULTS

3.1. *Results*

3.1.1. *Band 3 Continuum*

The Band 3 continuum image (Figure 1) resolved compact emission at the nucleus of NGC4429 with a synthesized beam of 1.63×1.55 arcsec at position angle 80.4 deg and an RMS noise of 0.0440 mJy/beam. We measured aperture photometry at three radii centered on the nucleus, with all three apertures yielding detections. Within an aperture of radius 1 beam (1.84 arcsec, corresponding to 147 pc at our adopted distance of 16.5 Mpc), we measured a flux density of 0.927 ± 0.084 mJy (S/N = 11.0). Enlarging the aperture to a radius of 2 beams (3.68 arcsec = 295 pc) gave 1.069 ± 0.168 mJy (S/N

= 6.4), while the 3-beam aperture (5.52 arcsec = 442 pc) gave 0.852 ± 0.254 mJy (S/N = 3.4). The flux did not grow appreciably beyond the 1-beam aperture, and the apparent decrease at 3 beams was consistent, within the quoted uncertainties, with the emission being dominated by a compact nuclear source with little extended contribution. The three photometry apertures are overlaid on the continuum image in Figure 1.

3.1.2. HCN(1-0) Emission

We searched the HCN(1-0) cube (rest frequency 88.6316 GHz) over ± 300 km/s (13 channels of 49.5 km/s width) around the adopted heliocentric systemic velocity of 1104 km/s. The cube had a synthesized beam of 1.88×1.81 arcsec at position angle 79.8 deg and a per-channel RMS of 0.848 mJy/beam. No significant integrated HCN(1-0) emission was detected in any of the three apertures; we therefore reported 3-sigma upper limits throughout. In the 1-beam aperture (1.84 arcsec = 147 pc) the integrated flux was 0.765 ± 0.255 Jy km/s (S/N = 1.47), corresponding to $L'_{\text{HCN}} = 8.684 \times 10^5$ K km/s pc² and, adopting $\alpha_{\text{HCN}} = 10.0$ Msun / (K km/s pc²) (Gao & Solomon 2004), a dense-gas mass of 8.684×10^6 Msun. The 2-beam aperture (3.68 arcsec = 295 pc) gave 1.515 ± 0.505 Jy km/s (S/N = 2.23), $L'_{\text{HCN}} = 1.720 \times 10^6$ K km/s pc², and a dense-gas mass of 1.720×10^7 Msun. The 3-beam aperture (5.52 arcsec = 442 pc) gave 2.261 ± 0.754 Jy km/s (S/N = 1.86), $L'_{\text{HCN}} = 2.566 \times 10^6$ K km/s pc², and a dense-gas mass of 2.566×10^7 Msun. All of these luminosities and masses should be read as 3-sigma upper limits, since none of the underlying integrated fluxes constituted a detection. The HCN(1-0) moment 0, 1, and 2 maps are shown in Figure 2.

3.1.3. Central-Depression Metric

To test for a central deficit in dense gas, we compared the 1-beam aperture flux to the flux in the annulus between 1 and 3 beam radii, which had a 3-sigma upper limit of 2.128 Jy km/s. The log10 ratio of the 1-beam flux to the 1-3 beam annulus flux was -0.444. However, both the central (1-beam) aperture and the 1-3 beam annulus were non-detections at the achieved sensitivity: the data cannot distinguish a central deficit from an overall HCN(1-0)-faint nucleus, and this metric was therefore inconclusive given the depth of the limits.

3.1.4. Kinematics

In the moment-1 map, 383 spatial pixels exceeded the 3-sigma per-channel threshold, spanning a velocity range of 804 to 1399 km/s about the systemic velocity of 1104 km/s. We noted, however, that this velocity range corresponds closely to the full ± 300 km/s

search window, and that the aperture-integrated fluxes at these positions did not constitute significant detections. The position-velocity cut through the nucleus is shown in Figure 3. Given the marginal per-pixel significance and the non-detection of integrated emission, we did not consider these data sufficient to establish the presence or absence of ordered rotation in the dense gas; the moment-1 velocity spread was consistent with individual low-significance channel peaks distributed across the searched window.

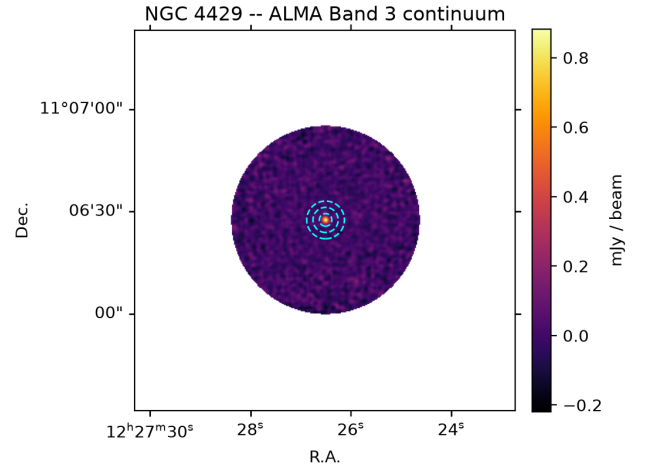


Figure 1. ALMA continuum image with the beam-scaled photometry apertures overlaid.

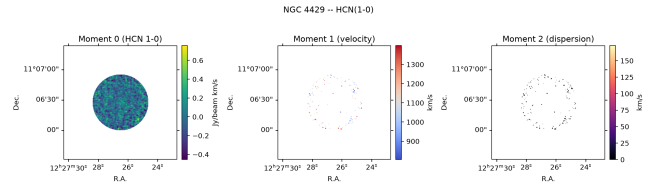


Figure 2. HCN(1-0) moment 0 (integrated intensity), moment 1 (velocity), and moment 2 (velocity dispersion) maps.

4. CONCLUSIONS

4.1. Conclusions

We have presented an HCN(1-0) and 3 mm continuum study of the circumnuclear region of NGC 4429, based on two ALMA Band 3 science-ready products and executed under a pre-registered analysis plan. Our principal conclusions are as follows.

First, HCN(1-0) is not detected anywhere in the inner few hundred parsecs of NGC 4429 at the achieved sen-

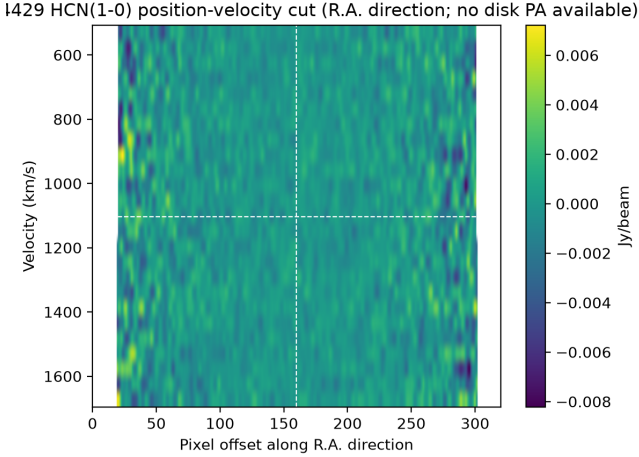


Figure 3. Position-velocity cut through the nucleus.

sitivity. With a per-channel RMS of 0.848 mJy/beam and a ± 300 km/s search window around the systemic velocity of 1104 km/s, the integrated fluxes in nested apertures of one, two, and three beam radii (147, 295, and 442 pc) all fall below the pre-registered detection criterion, at signal-to-noise ratios of 1.5–2.2. The resulting 3σ upper limits — 0.765, 1.515, and 2.261 Jy km/s — translate, for $\alpha_{\text{HCN}} = 10 \text{ M}_{\text{sun}} (\text{K km/s pc}^2)^{-1}$, into dense-gas mass ceilings of 8.7×10^6 , 1.7×10^7 , and $2.6 \times 10^7 \text{ M}_{\text{sun}}$. These are the first HCN(1-0)-based limits on the dense-gas reservoir around the black hole in NGC 4429, and they carry the usual factor-of-several systematic from the conversion factor, which we quote as a bracket rather than fold into the statistical error. The tentative velocity structure seen in individual low-significance pixels spans essentially the full search window and does not survive aperture integration; we draw no kinematic conclusions from it.

Second, and central to the motivating question: the status of the reported central molecular hole remains unconstrained by these data. The pre-defined central-depression metric — the logarithmic ratio of the central-beam flux to the one-to-three-beam annulus flux, measured at -0.444 — compares two non-detections, and our beam-convolved disk models show that no cavity configuration in the bracketed family can be excluded under every profile assumption at this depth. A nucleus with a genuine dense-gas cavity and a nucleus that is uniformly faint in HCN(1-0) produce indistinguishable signatures at this sensitivity and resolution. We emphasize that “unconstrained” is the honest verdict here, and that the mass limits above — not any statement about the cavity — are the quantitative deliverable of the line analysis.

Third, the nucleus is unambiguously detected in the 3 mm continuum, at $0.927 \pm 0.084 \text{ mJy}$ ($S/N = 11$) in

the central beam, with no significant additional flux in larger apertures. The emission is therefore compact on the ~ 150 pc scale of the beam and coincident with the nuclear position, indicating a persistent compact source — plausibly low-level nuclear activity — at the center of the CO-defined hole. Whatever process cleared or dimmed the molecular tracers at the nucleus, it has not left the nucleus dark at 3 mm.

Two limitations of the input data dominate the interpretation and should be stated plainly. The first is angular resolution: the synthesized beams ($1.63''$ – $1.88''$, i.e., $\gtrsim 130$ – 150 pc at 16.5 Mpc) are large compared with the tens-of-parsecs scale of the reported CO hole, so the central-beam measurement necessarily blends any cavity with the surrounding disk, and beam convolution alone can erase the signature of even a total central deficit. The second is spectral coverage and channelization: the delivered cube is a representative-bandwidth product with 49.5 km/s channels, giving only 13 channels across our ± 300 km/s search window. This coarse velocity sampling dilutes narrow lines within each channel, leaves little line-free baseline margin around the expected emission, and rules out any meaningful kinematic analysis — a limitation we anticipated by declining to compute moment-2 maps or fit rotation.

These limitations point directly to what would settle the question. A deeper, higher-resolution HCN(1-0) (or HCO+(1-0)) observation with a synthesized beam of $\lesssim 0.5''$ — a few tens of parsecs, comparable to the hole itself — would allow the cavity to be resolved rather than modeled, and would need to reach a per-channel sensitivity several times better than the 0.848 mJy/beam achieved here to detect disk-level dense-gas emission or place limits deep enough to exclude filled-disk models. Equally important, such an observation should use a dedicated spectral setup with fine channelization ($\lesssim 10$ – 20 km/s) over a window comfortably wider than the disk’s projected rotation, so that line profiles can be resolved and velocity-coherent stacking becomes possible. Alternatively, higher-J HCN transitions at correspondingly finer native resolution, or a matched-resolution comparison of HCN against existing high-resolution CO imaging, would allow the dense-gas fraction to be mapped across the hole boundary — the measurement this study could not defensibly make at mismatched resolutions. Until such data exist, the appropriate summary of the NGC 4429 molecular hole is the one this study was designed to be able to give: the dense-gas mass within ~ 150 pc of the black hole is below $\sim 10^7 \text{ M}_{\text{sun}}$ for standard conversion factors, the 3 mm nucleus is active and compact, and the emptiness of the hole itself remains an open question with a now

well-defined sensitivity and resolution target for closing it.

5. SOFTWARE AND DATA

This paper’s Abstract, Introduction, Methods, Results, and Conclusions text was drafted by an LLM-based automated pipeline, conditioned strictly on real measurements from the analysis code described in Methods – no simulated or placeholder numbers appear in the Results section. This is a pilot/demonstration output and has not been independently peer reviewed; a domain expert should verify every quantitative claim before any further use.

6. REFERENCES

1. Gao, Y. & Solomon, P. M. 2004, ApJS, 152, 63 (dense-gas mass conversion factor, `alpha_HCN`)
2. Adopted distance (16.5 Mpc) and systemic velocity (1104.0 km/s): The data description states neither value, so both come from the literature: 16.5 Mpc is the surface-brightness-fluctuation-based distance adopted for NGC 4429 in the ATLAS3D survey (Cappellari et al. 2011) and reused in the WISDOM CO dynamical black-hole measurement of this galaxy (Davis et al. 2018, MNRAS), which is the standard reference for its SMBH. The heliocentric systemic velocity of ~ 1104 km/s likewise comes from ATLAS3D/NED and matches the value fitted in the Davis et al. 2018 gas-dynamical model; both are well-established, but note the velocity is quoted from memory of those catalogs rather than re-queried live here.